PVsyst - Simulation report

Standalone system

Project: WAJIR

Variant: New simulation

Standalone system with batteries

System power: 77.4 kWp

Wajir - Kenya

PVsyst V8.0.6
VC0, Simulation date:
01/09/26 15:00
with V8.0.6

Project summary

Geographical Site
Wajir
Kenya

Situation

Latitude	1.75 °N
Longitude	40.02 °E
Altitude	252 m
Time zone	UTC+3

Project settings

Albedo	0.20

Weather data
Wajir
Meteonorm 8.2 (1991-2007), Sat=100% - Synthetic

System summary

Standalone system

Standalone system with batteries

Orientation #1
Fixed plane

Tilt/Azimuth	20 / 0 °

User's needs
Daily household consumers
Constant over the year
Average 287 kWh/Day

System information

PV Array

Nb. of modules	120 units
Pnom total	77.4 kWp

Battery pack

Technology	Lithium-ion, LFP
Nb. of units	8 units
Voltage	410 V
Capacity	800 Ah

Results summary

Useful energy from solar	103.38 MWh/year	Specific production	1336 kWh/kWp/year	Perf. Ratio PR	60.31 %
Missing Energy	1.31 MWh/year	Available solar energy	140.69 MWh/year	Solar Fraction SF	98.74 %
Excess (unused)	34.27 MWh/year				

Table of contents

Project and results summary 2
General parameters, PV Array Characteristics, System losses 3
Detailed User's needs 5
Main results 6
Loss diagram 7
Predef. graphs 8
Cost of the system 13
Financial analysis 14

PVsyst V8.0.6
VC0, Simulation date:
01/09/26 15:00
with V8.0.6

General parameters

Standalone system

Standalone system with batteries

Orientation #1
Fixed plane
Tilt/Azimuth 20 / 0 °

Sheds configuration
No 3D scene defined

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

User's needs
Daily household consumers
Constant over the year
Average 287 kWh/Day

PV Array Characteristics

PV module
Manufacturer Jinkosolar
Model JKM-645N-66HL4-BDV
(Original PVsyst database)
Unit Nom. Power 645 Wp
Number of PV modules 120 units
Nominal (STC) 77.4 kWp
Modules 20 string x 6 In series
At operating cond. (50°C)
Pmpp 71.8 kWp
U mpp 230 V
I mpp 312 A

Controller
Universal controller
Technology MPPT converter
Temp coeff. -5.0 mV/°C/Elem.
Converter
Maxi and EURO efficiencies 97.0 / 95.0 %
Battery Management control
Threshold commands as SOC calculation
Charging SOC = 0.96 / 0.80
Discharging SOC = 0.10 / 0.35

Battery
Manufacturer Deye
Model BOS-G 8 - 40.96 kWh
Technology Lithium-ion, LFP
Nb. of units 8 in parallel
Discharging min. SOC 10.0 %
Stored energy 294.9 kWh
Battery Pack Characteristics
Voltage 410 V
Nominal Capacity 800 Ah (C10)
Temperature Fixed 20 °C

Total PV power
Nominal (STC) 77 kWp
Total 120 modules
Module area 324 m²
Cell area 303 m²

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 12 mΩ
Loss Fraction 1.5 % at STC

Serie Diode Loss
Voltage drop 0.7 V
Loss Fraction 0.3 % at STC

Module Quality Loss
Loss Fraction -0.8 %

Module mismatch losses
Loss Fraction 2.0 % at MPP

Strings Mismatch loss
Loss Fraction 0.2 %

PVsyst V8.0.6
VC0, Simulation date:
01/09/26 15:00
with V8.0.6

Array losses

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

PVsyst V8.0.6
VC0, Simulation date:
01/09/26 15:00
with V8.0.6

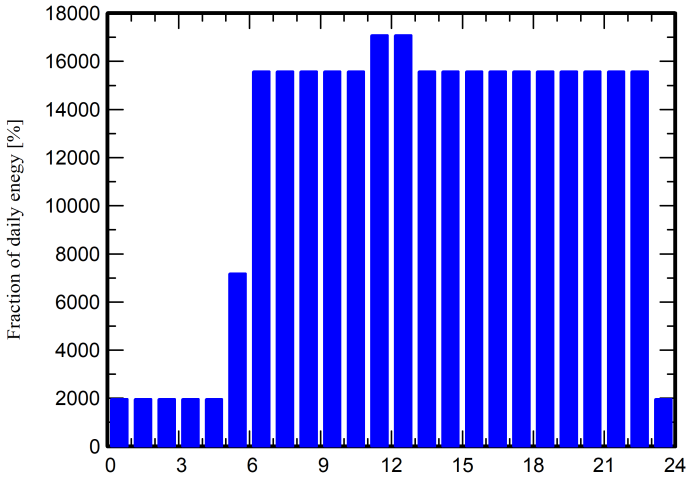
Detailed User's needs

Daily household consumers, Constant over the year, average = 287 kWh/day

Annual values

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	57	100/lamp	17.0	96900
Drink Fridges	6	870/app	18.0	93960
Wall mounted fans	35	60/app	17.0	35700
Deep-freezers	3		24	41040
Cashier Computers	4		17	10200
Accessories	1	250 tot	24.0	6000
Water Pump	1	1500 tot	2.0	3000
Stand-by consumers			24.0	24
Total daily energy				286824

Hourly distribution



Main results

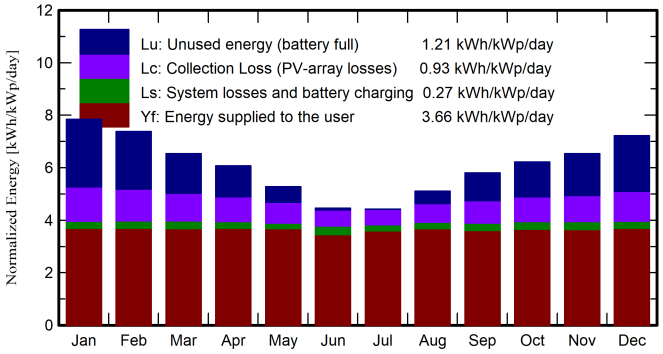
System Production

Useful energy from solar	103.38 MWh/year	Perf. Ratio PR	60.31 %
Available solar energy	140.69 MWh/year	Solar Fraction SF	98.74 %
Excess (unused)	34.27 MWh/year		
Loss of Load		**Battery aging (State of Wear)**	
Time Fraction	1.8 %	Cycles SOW	96.2 %
Missing Energy	1.31 MWh/year	Static SOW	90.0 %
		Battery lifetime	10.0 years

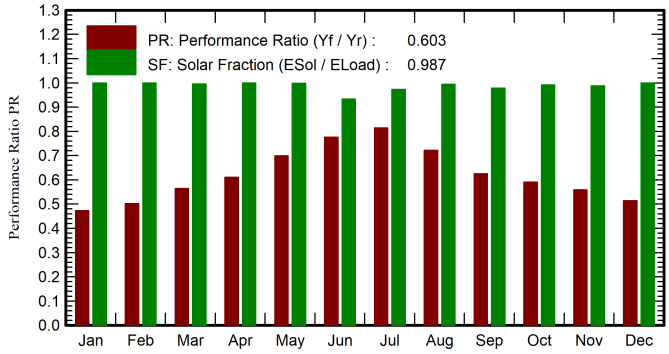
Economic evaluation

Investment		**Yearly cost**		**LCOE**	
Global	5,679,160.00 Ksh	Annuities	0.00 Ksh/yr	Energy cost	2.13 Ksh/kWh
Specific	73.4 Ksh/Wp	Run. costs	503,381.61 Ksh/yr		
		Payback period	2.7 years		

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

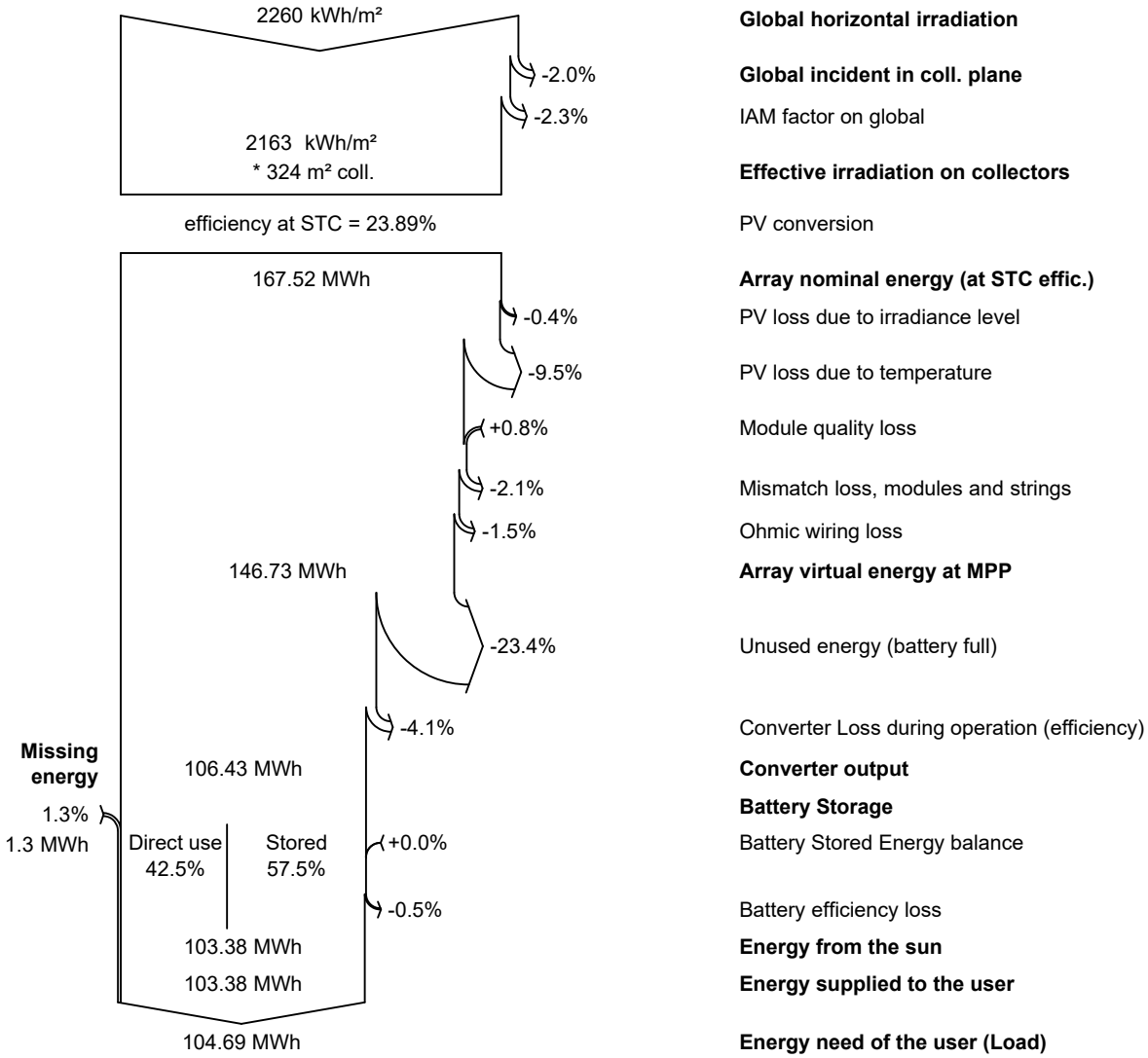
	GlobHor kWh/m²	GlobEff kWh/m²	E_Avail MWh	EUnused MWh	E_Miss MWh	E_User MWh	E_Load MWh	SolFrac ratio
January	215.0	239.6	15.31	6.16	0.000	8.89	8.89	1.000
February	194.2	203.5	13.03	4.74	0.000	8.03	8.03	1.000
March	206.7	198.2	12.77	3.59	0.042	8.85	8.89	0.995
April	202.3	177.4	11.54	2.72	0.000	8.60	8.60	1.000
May	194.8	158.1	10.39	1.43	0.021	8.87	8.89	0.998
June	161.8	128.7	8.56	0.14	0.575	8.03	8.60	0.933
July	162.0	132.2	8.84	0.00	0.244	8.65	8.89	0.973
August	178.5	153.4	10.19	1.11	0.048	8.84	8.89	0.995
September	182.8	169.9	11.09	2.44	0.192	8.41	8.60	0.978
October	187.3	189.0	12.24	3.16	0.079	8.81	8.89	0.991
November	179.1	192.8	12.50	3.71	0.113	8.49	8.60	0.987
December	195.7	220.4	14.21	5.07	0.000	8.89	8.89	1.000
Year	2260.0	2163.4	140.69	34.27	1.314	103.38	104.69	0.987

Legends

GlobHor	Global horizontal irradiation	E_User	Energy supplied to the user
GlobEff	Effective Global, corr. for IAM and shadings	E_Load	Energy need of the user (Load)
E_Avail	Available Solar Energy	SolFrac	Solar fraction (EUsed / ELoad)
EUnused	Unused energy (battery full)		
E_Miss	Missing energy		

PVsyst V8.0.6
VC0, Simulation date:
01/09/26 15:00
with V8.0.6

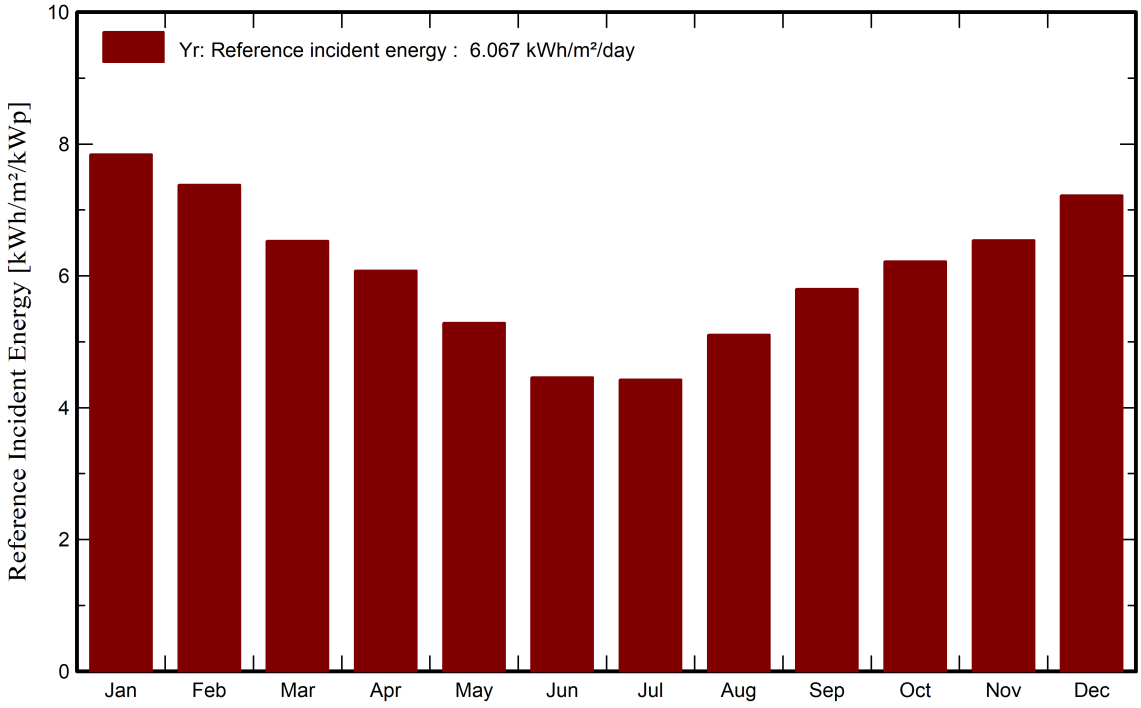
Loss diagram



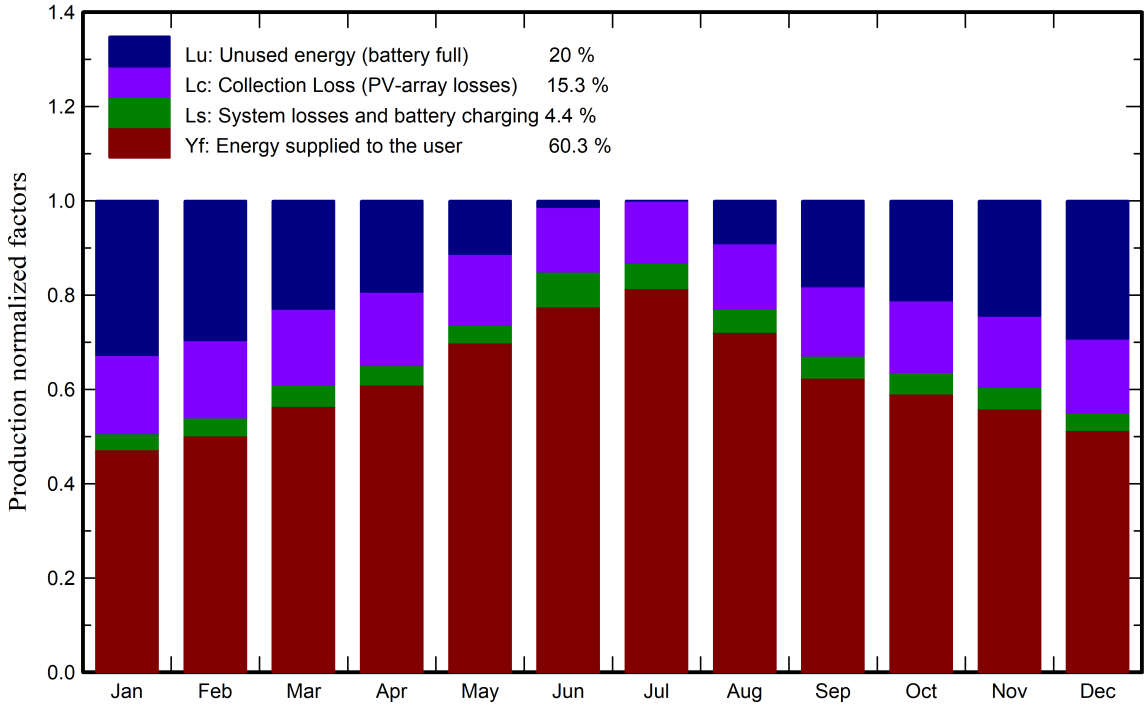
PVsyst V8.0.6
VC0, Simulation date:
01/09/26 15:00
with V8.0.6

Predef. graphs

Reference Incident Energy in Collector Plane

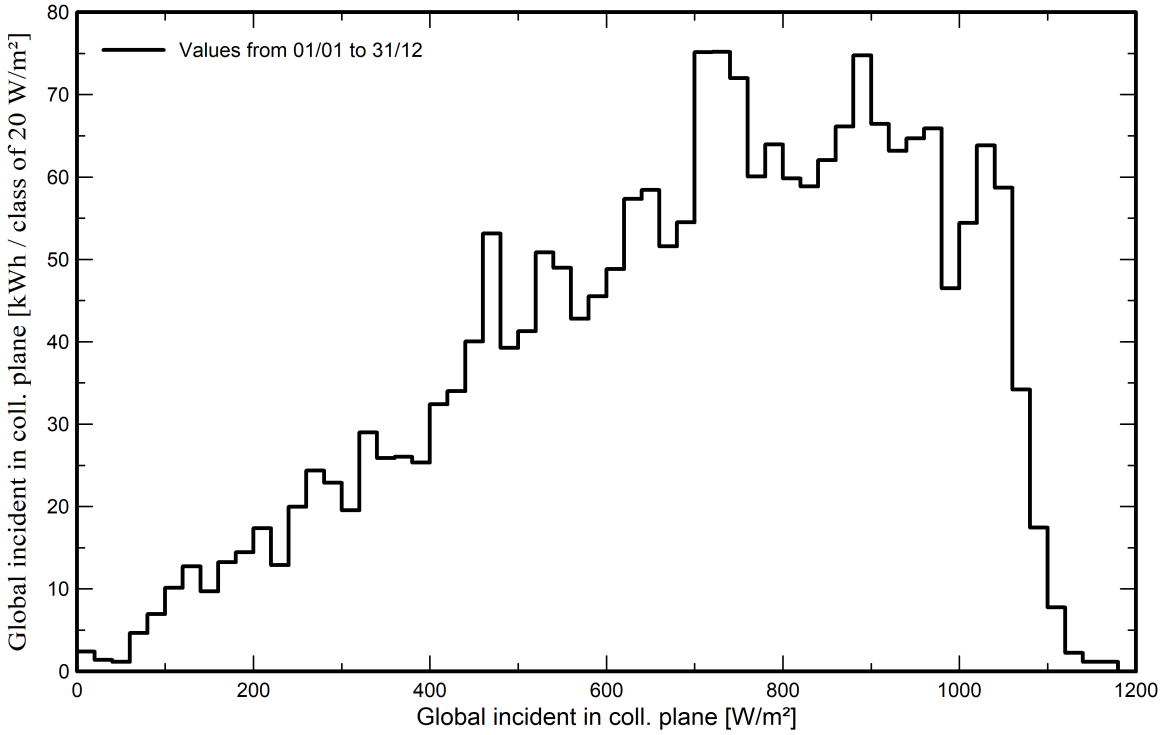


Normalized Production and Loss Factors

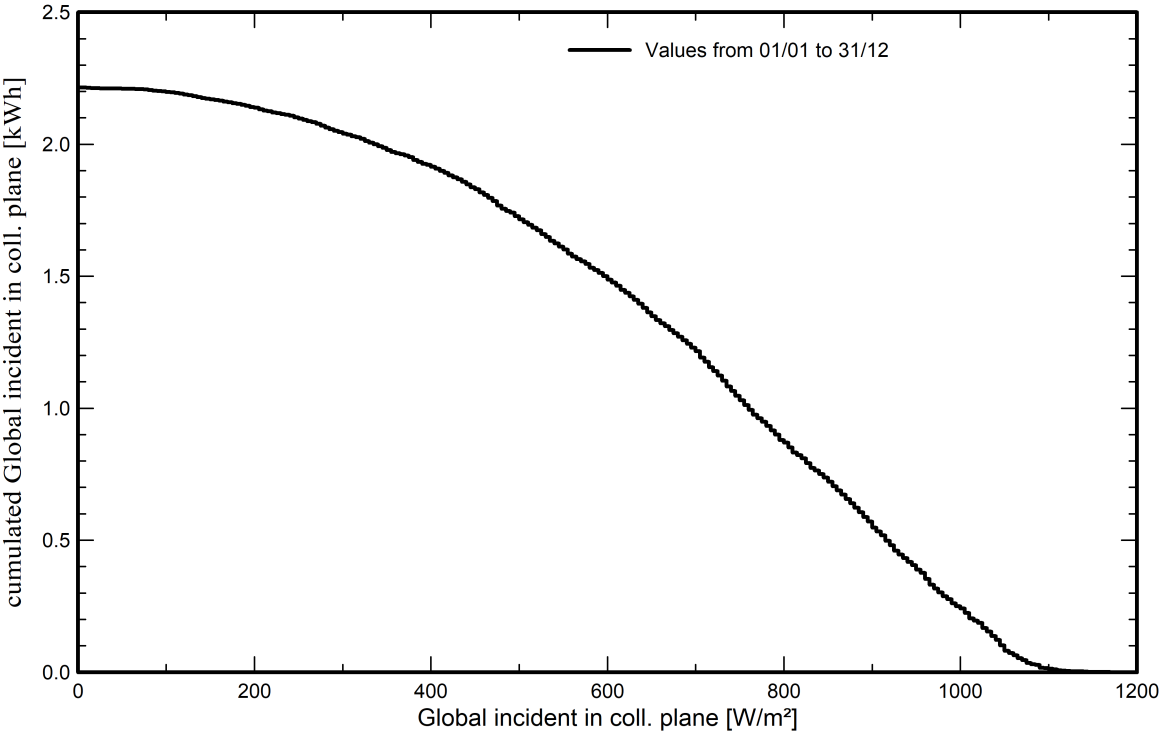


Predef. graphs

Incident Irradiation Distribution

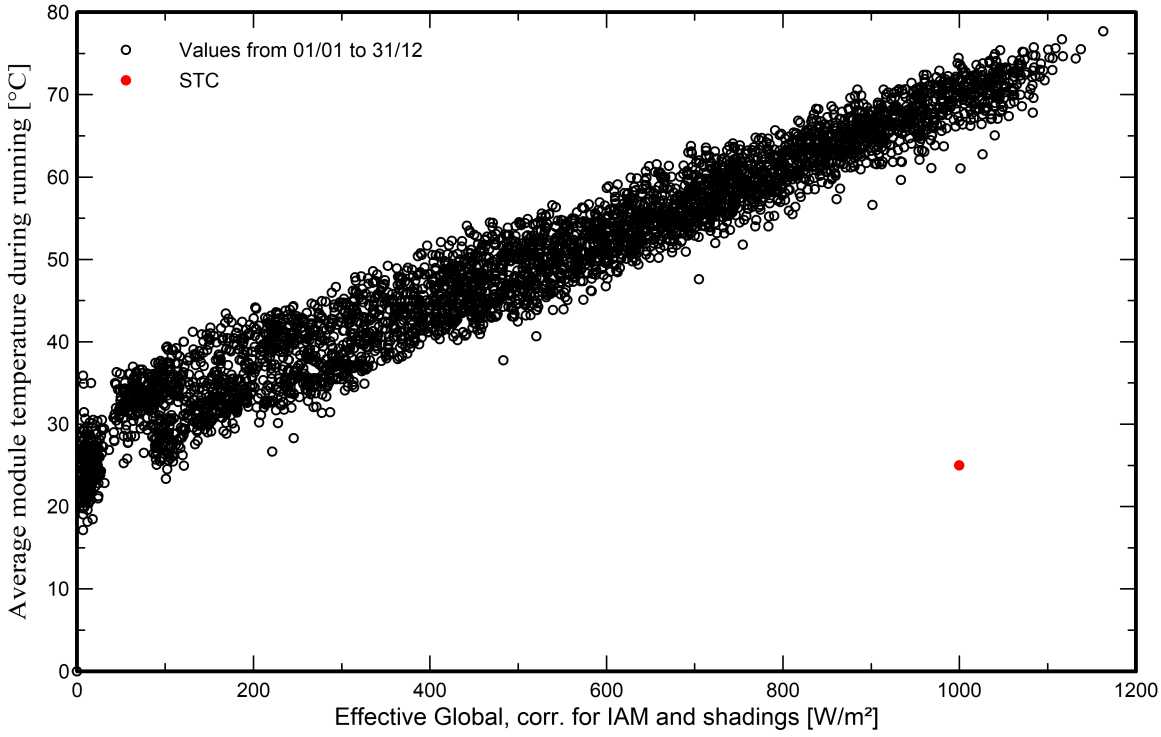


Incident Irradiation cumulative distribution

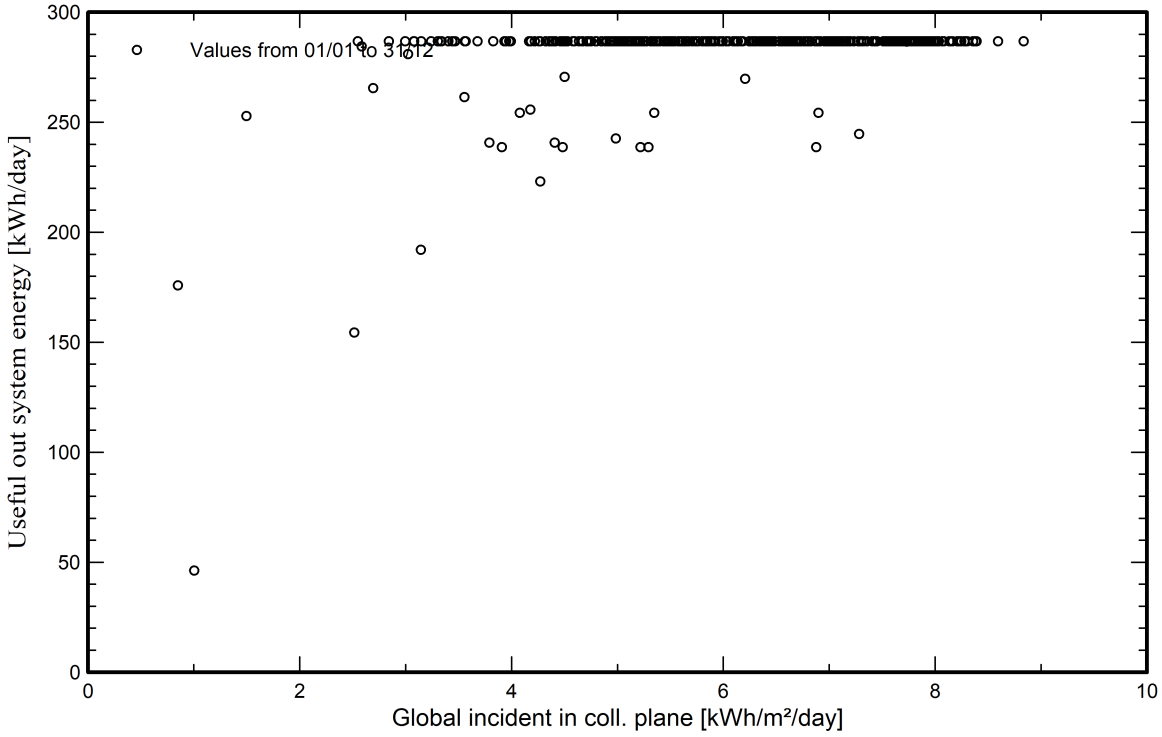


Predef. graphs

Array Temperature vs. Effective Irradiance



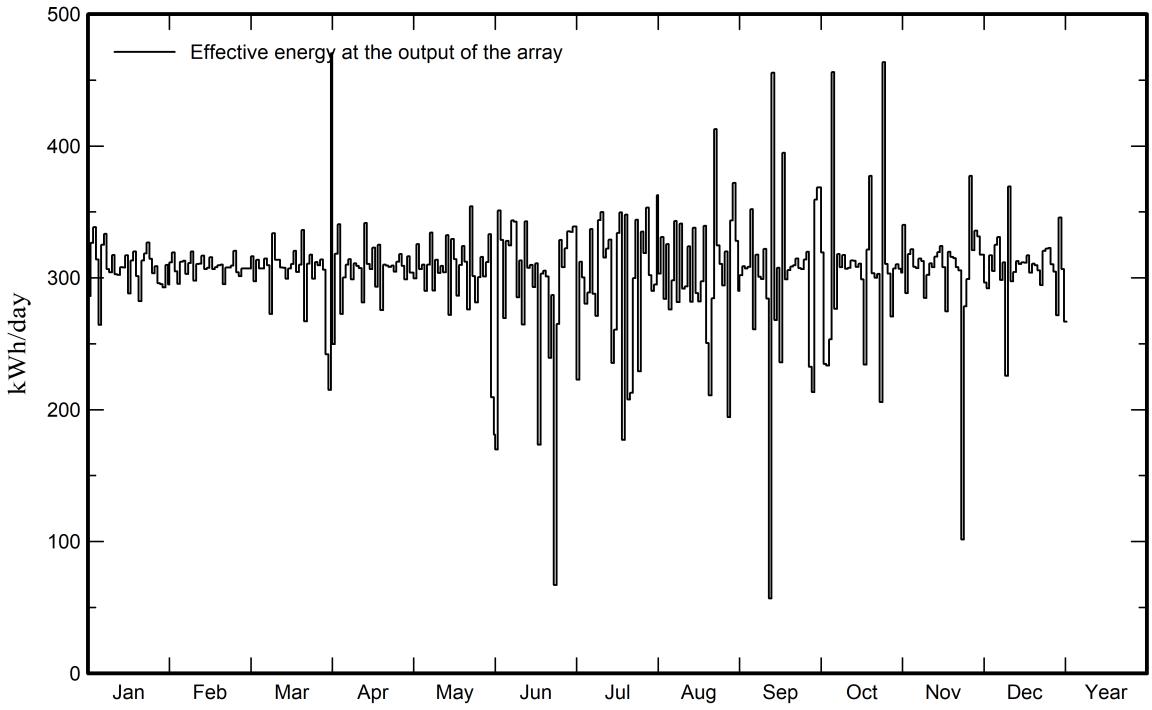
Daily Input/Output diagram



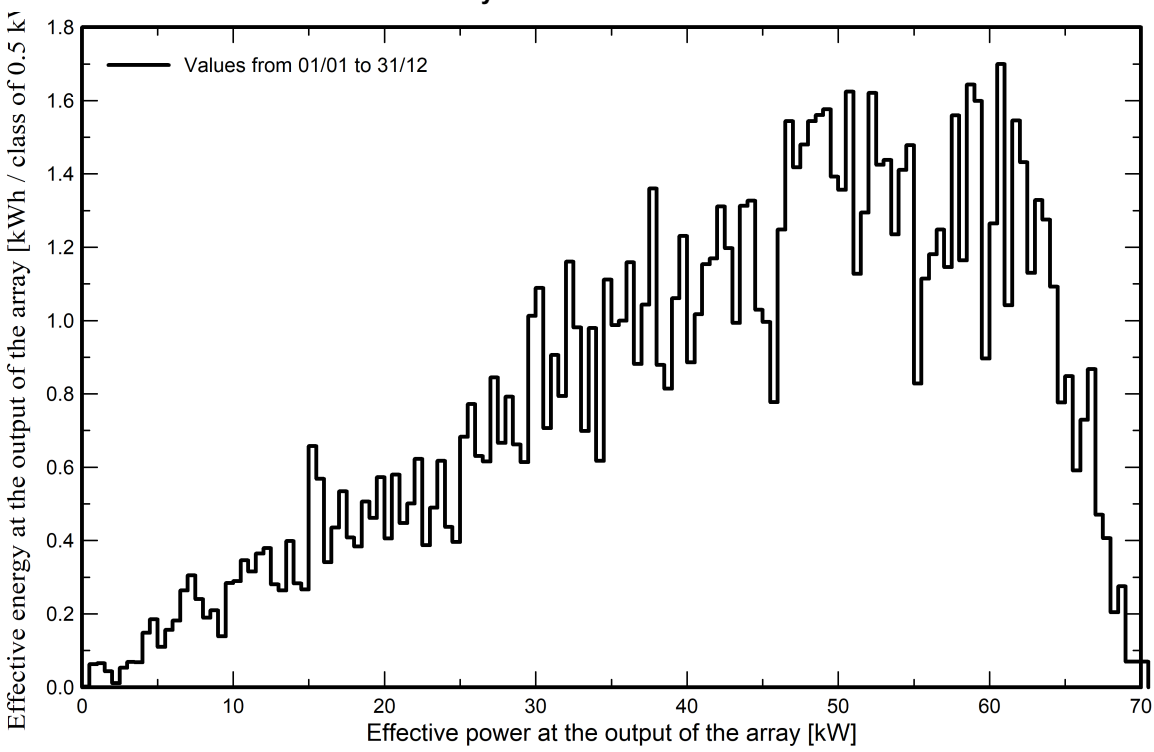
PVsyst V8.0.6
VC0, Simulation date:
01/09/26 15:00
with V8.0.6

Predef. graphs

Daily Array Output Energy



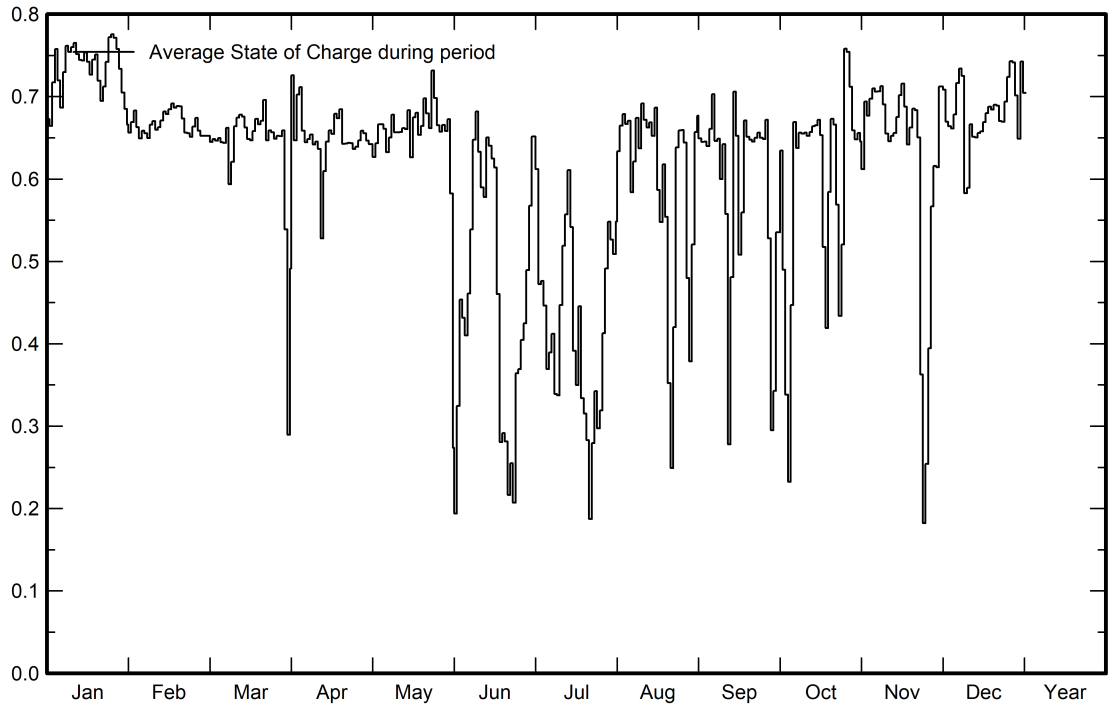
Array Power Distribution



PVsyst V8.0.6
VC0, Simulation date:
01/09/26 15:00
with V8.0.6

Predef. graphs

State of charge daily distribution



Cost of the system

Installation costs

Item	Quantity units	Cost Ksh	Total Ksh
PV modules			
JKM-645N-66HL4-BDV	120	13,000.00	1,560,000.00
Supports for modules	100	500.00	50,000.00
Batteries	8	300,000.00	2,400,000.00
Controllers			35,000.00
Inverter	2	700,000.00	1,400,000.00
Other components			
Accessories, fasteners	2500	1.00	2,500.00
Wiring	40000	1.00	40,000.00
Combiner box	20000	1.00	20,000.00
Monitoring system, display screen	5000	1.00	5,000.00
Measurement system, pyranometer	5000	1.00	5,000.00
Surge arrester	2500	1.00	2,500.00
Studies and analysis			
Engineering	1	2,500.00	2,500.00
Permitting and other admin. Fees	1	2,500.00	2,500.00
Environmental studies	1	2,500.00	2,500.00
Economic analysis	1	2,500.00	2,500.00
Installation			
Global installation cost per module	120	750.00	90,000.00
Global installation cost per controller	1	1,000.00	1,000.00
Global installation cost per battery	8	2,500.00	20,000.00
Global Installation cost per Inverter	2	2,500.00	5,000.00
Settings	1	5,000.00	5,000.00
Taxes			
VAT	1	0.00	28,160.00
		Total	5,679,160.00
		Depreciable asset	4,047,500.00

Operating costs

Item	Total Ksh/year
Maintenance	
Repairs	150,000.00
Cleaning	20,000.00
Security fund	50,000.00
Total (OPEX)	220,000.00
Including inflation (8.00%)	503,381.61

System summary

Total installation cost	5,679,160.00	Ksh
Operating costs (incl. inflation 8.00%/year)	503,381.61	Ksh/year
Excess energy (battery full)	34.3	MWh/year
Used solar energy	103	MWh/year
Used energy cost	7.6162	Ksh/kWh

Financial analysis

Simulation period

Project lifetime	20 years	Start year	2027

Income variation over time

Inflation	8.00 %/year
Production variation (aging)	0.00 %/year
Discount rate	0.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (Ksh)	Depreciable (Ksh)
PV modules				
JKM-645N-66HL4-BDV	Straight-line	20	0.00	1,560,000.00
Supports for modules	Straight-line	20	0.00	50,000.00
Batteries	Straight-line	20	0.00	2,400,000.00
Controllers	Straight-line	20	0.00	35,000.00
Accessories, fasteners	Straight-line	20	0.00	2,500.00
		Total	0.00	4,047,500.00

Financing

Own funds	5,679,160.00 Ksh

Electricity sale

Feed-in tariff	25.00000 Ksh/kWh

Return on investment

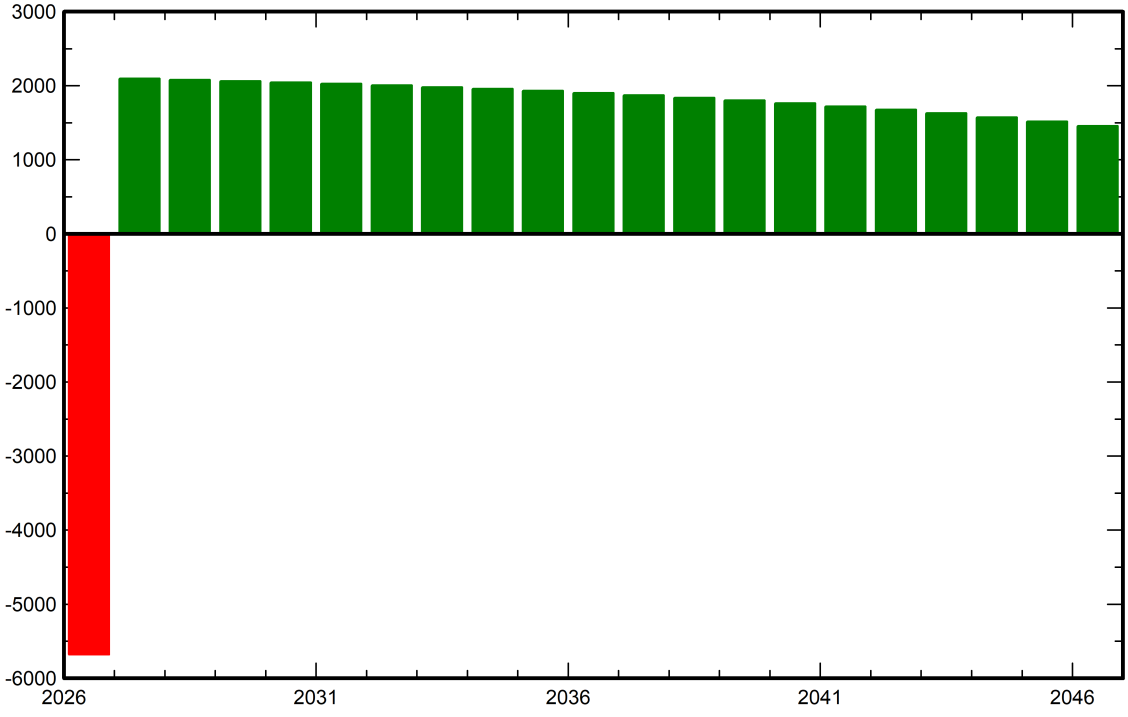
Payback period	2.7 years
Net present value (NPV)	31,244,997.13 Ksh
Internal rate of return (IRR)	35.86 %
Return on investment (ROI)	550.2 %

Financial analysis

Detailed economic results (kKsh)

Year	Electricity sale	Own funds	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Cumul. profit	% amorti.
0	0	5,679,160	0	0	0	0	0	-5,679,160	0.0%
1	2,584,423	0	220,000	202,375	2,162,048	270,256	2,094,167	-3,584,993	36.9%
2	2,584,423	0	237,600	202,375	2,144,448	268,056	2,078,767	-1,506,226	73.5%
3	2,584,423	0	256,608	202,375	2,125,440	265,680	2,062,135	555,908	109.8%
4	2,584,423	0	277,137	202,375	2,104,911	263,114	2,044,172	2,600,080	145.8%
5	2,584,423	0	299,308	202,375	2,082,740	260,343	2,024,773	4,624,853	181.4%
6	2,584,423	0	323,252	202,375	2,058,796	257,349	2,003,821	6,628,674	216.7%
7	2,584,423	0	349,112	202,375	2,032,935	254,117	1,981,193	8,609,868	251.6%
8	2,584,423	0	377,041	202,375	2,005,006	250,626	1,956,756	10,566,623	286.1%
9	2,584,423	0	407,205	202,375	1,974,843	246,855	1,930,363	12,496,986	320.0%
10	2,584,423	0	439,781	202,375	1,942,267	242,783	1,901,858	14,398,844	353.5%
11	2,584,423	0	474,963	202,375	1,907,084	238,386	1,871,074	16,269,918	386.5%
12	2,584,423	0	512,961	202,375	1,869,087	233,636	1,837,826	18,107,744	418.8%
13	2,584,423	0	553,997	202,375	1,828,050	228,506	1,801,919	19,909,663	450.6%
14	2,584,423	0	598,317	202,375	1,783,731	222,966	1,763,139	21,672,803	481.6%
15	2,584,423	0	646,183	202,375	1,735,865	216,983	1,721,257	23,394,060	511.9%
16	2,584,423	0	697,877	202,375	1,684,171	210,521	1,676,024	25,070,084	541.4%
17	2,584,423	0	753,707	202,375	1,628,340	203,543	1,627,173	26,697,257	570.1%
18	2,584,423	0	814,004	202,375	1,568,044	196,005	1,574,413	28,271,670	597.8%
19	2,584,423	0	879,124	202,375	1,502,923	187,865	1,517,433	29,789,103	624.5%
20	2,584,423	0	949,454	202,375	1,432,593	179,074	1,455,894	31,244,997	650.2%
Total	**51,688,455**	**5,679,160**	**10,067,632**	**4,047,500**	**37,573,322**	**4,696,665**	**36,924,157**	**31,244,997**	**650.2%**

Yearly net profit (kKsh)



PVsyst V8.0.6
VC0, Simulation date:
01/09/26 15:00
with V8.0.6

Financial analysis

Cumulative cashflow (kKsh)

